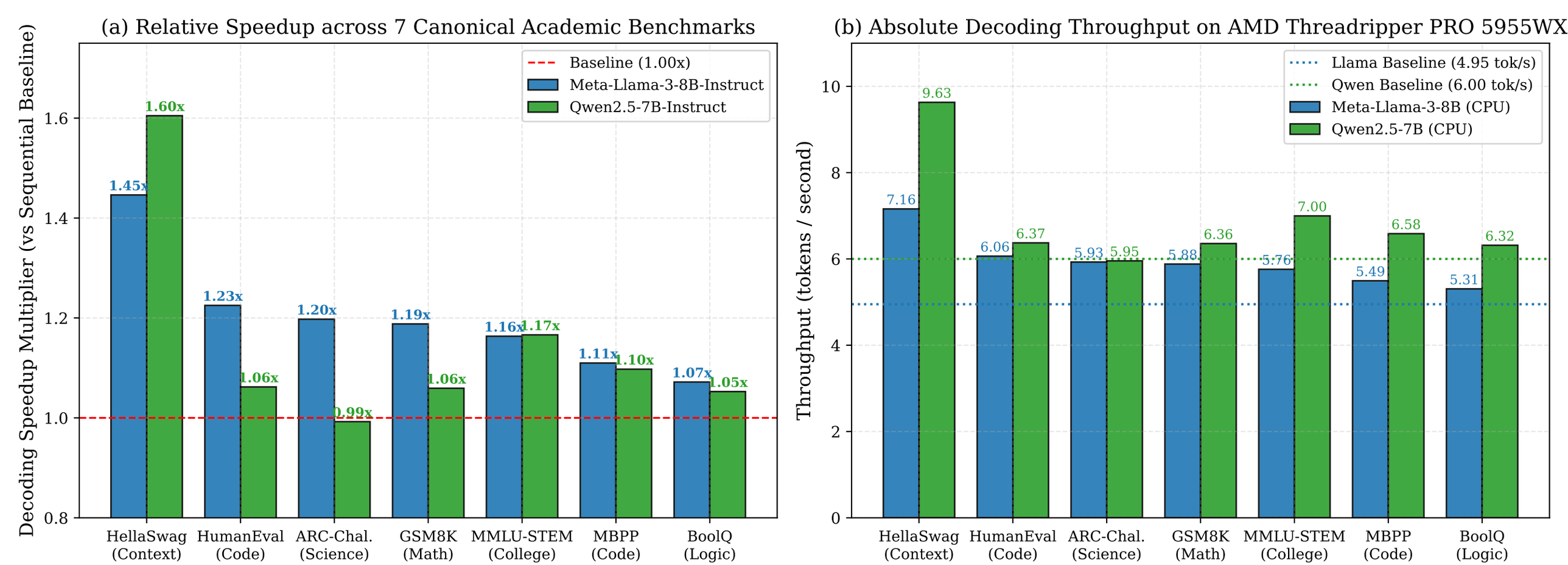
(a) Relative Speedup across 7 Canonical Academic Benchmarks
Decoding Speedup Multiplier (vs Sequential Baseline)
Baseline (1.00x)
Meta-Llama-3-8B-Instruct
Qwen2.5-7B-Instruct
1.45x
1.60x
1.23x
1.06x
1.20x
0.99x
1.19x
1.06x
1.16x
1.17x
1.11x
1.10x
1.07x
1.05x
HellaSwag (Context)
HumanEval (Code)
ARC-Chal. (Science)
GSM8K (Math)
MMLU-STEM (College)
MBPP (Code)
BoolQ (Logic)
(b) Absolute Decoding Throughput on AMD Threadripper PRO 5955WX
Throughput (tokens / second)
Llama Baseline (4.95 tok/s)
Qwen Baseline (6.00 tok/s)
Meta-Llama-3-8B (CPU)
Qwen2.5-7B (CPU)
7.16
9.63
6.06
6.37
5.93
5.95
5.88
6.36
5.76
7.00
5.49
6.58
5.31
6.32
HellaSwag (Context)
HumanEval (Code)
ARC-Chal. (Science)
GSM8K (Math)
MMLU-STEM (College)
MBPP (Code)
BoolQ (Logic)